

From Evaluation to Improvement: Boundary-Guided Training of Closed-Loop Controllers

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Abstract

A frozen controller’s failures across operating conditions form a map of where it works. We propose to investigate whether training near an estimated performance boundary expands that region while limiting regression where the controller was already acceptable. We certify the region of a frozen policy, feed that estimate into the choice of training conditions, and re-certify on fresh data. Monotonic improvement is the ambitious target, not an assumption.

1. Motivation, Problem Formulation, and Feedback Loops

Evaluating a frozen controller across operating conditions maps where it works, not just how well. On CartPole with the pole-angle cutoff relaxed to 90° , call an episode a *recovery* if it survives 500 steps with final-100-step RMS pole angle at most 5° . An adaptive sweep of initial pole angles and half-lengths (248 conditions, 50 seeds, 37,200 episodes, no retraining) gives Figure 1: quantized LQR and PPO have recovery regions differing in shape, not only size, and survival and recovery disagree, since antisymmetrized PPO often survives from $\pm 25^\circ$ without settling inside the band. We ask whether evaluating that boundary can guide training that *expands* the acceptable region while limiting degradation elsewhere, which signal—threshold proximity, intermediate difficulty, uncertainty, or improbability—marks a useful training condition, and what assumptions permit a guarantee. Informative *evaluation* conditions need not be informative *training* conditions.

At iteration k , freeze π_k . For a condition $z \in \mathcal{Z}$, horizon H , and declared episode randomness $\xi \sim P_\xi(\cdot | z)$, let $p_k(z) = \Pr_\xi[\tau_{\pi_k}(z, \xi) \text{ fails recovery}]$; under tolerance α the acceptable region is $S_k = \{z : p_k(z) \leq \alpha\}$. With a measure μ fixed in advance, $V_k = \mu(S_k)$, $g_k = \mu(S_{k+1} \setminus S_k)$, and $\ell_k = \mu(S_k \setminus S_{k+1})$ satisfy $V_{k+1} - V_k = g_k - \ell_k$. Three loops nest—**control**: observation $\rightarrow$ action $\rightarrow$ next observation; **evaluation**: condition selection $\rightarrow$ frozen rollout $\rightarrow$ updated evidence $\rightarrow$ next condition; and **improvement**, studied here:

$$\pi_k \longrightarrow \hat{p}_k, \hat{S}_k \longrightarrow q_k = (1 - \lambda)q_{\text{base}} + \lambda q_{\text{target},k} \longrightarrow \text{Train}(\pi_k, q_k) \longrightarrow \pi_{k+1}.$$

Closing it moves the target from p_k to p_{k+1} ; policies change only in a separate training phase, so each $\hat{p}_k$ describes one closed-loop system. What is distinctively L4DC is that the estimated quantity is a closed-loop property of a learned controller over initial conditions and plant parameters, not a reward, and that the object to certify and grow is a region, not a scalar.

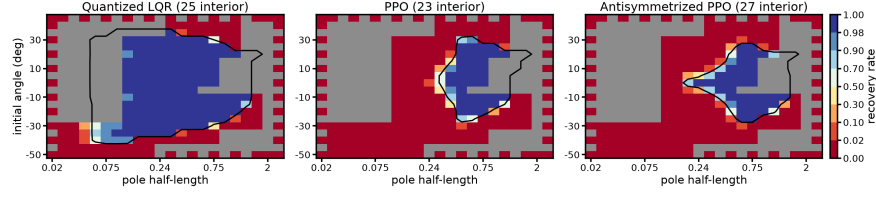


Figure 1: Preliminary recovery rate for three *frozen* controllers on the angle/length lattice in mesh-rank coordinates (248 of 440 slots sampled, gray unsampled; black line is the 0.5 contour). Over 92% of cells are exactly 0 or 1: the interior cells are the whole boundary.

2. Related Work

Evaluation. Gotovos et al. (2013) and Letham et al. (2022) develop adaptive level-set estimation, the latter for Bernoulli observations like ours; Howard et al. (2021) supply time-uniform confidence sequences, so adaptive sampling still certifies. **Training-condition selection.** Reverse curriculum generation (Florensa et al., 2017) adapts start states to current performance, prioritized level replay (Jiang et al., 2021) ranks configurations by learning potential, and sampling for learnability (Rutherford et al., 2024) targets mixed-success conditions, so boundary-guided sampling is not itself novel; our contribution is the controlled comparison under a fixed, certified protocol. **Control guarantees.** Berkenkamp et al. (2017) link model-based learning to safe-region expansion via Lyapunov analysis; their stability certificates differ from our finite-horizon recovery probabilities.

3. Goals

Stage 1 (course foundation). Build the evaluator: certify a condition only when a confidence sequence puts its failure rate strictly on one side of α , and split $\sum_k \delta_k \leq \delta$ so all certified sets hold with probability $1 - \delta$ (proved by Hoeffding and a union bound). For $x_{j+1} = (A - BK)x_j$ and $J_{H,K}(x) = x^\top P_H(K)x$ on $\|x\|_2 \leq r$, the update bound $|J_{H,K'}(x) - J_{H,K}(x)| \leq r^2 \|P_H(K') - P_H(K)\|_2 =: b$ confines all regression of the sublevel set $\{J_{H,K} \leq h\}$ to the band $h - b < J_{H,K} \leq h$. These elementary results do not transfer automatically to the nonlinear learned system.

Stage 2 (main research). From one frozen PPO checkpoint, branch into matched training runs under three selection rules—uniform, boundary-guided over a band at the estimated boundary, and intermediate-difficulty scored by $\hat{p}_k(1 - \hat{p}_k)$ —holding q_{base} , λ , the objective, and the optimizer fixed. Starting with initial-angle variation, then the angle–length domain of Figure 1, we report g_k , ℓ_k , net coverage, maps of *where* change occurs, seed variation, and total cost. Each updated policy is re-certified on fresh data, separating certificate growth from retraining. A negative result is a real outcome.

Stage 3 (ambitious target). Bound performance drift by update size, horizon, and closed-loop amplification; construct examples where improving selected conditions degrades others; and find a restricted controller class and update admitting a sufficient condition for expansion. Both the continuous-cost sublevel set and the level set S_k are candidates. **Open decisions.** Whether repeated rounds are required or a single $\pi_0 \rightarrow \pi_1$ update suffices, and which level-set notion is primary, stay open; α , δ , budgets, seeds, reward, and mixture weights will be fixed and recorded beforehand.

References

- Felix Berkenkamp, Matteo Turchetta, Angela P. Schoellig, and Andreas Krause. Safe model-based reinforcement learning with stability guarantees. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/766ebcd59621e305170616ba3d3dac32-Abstract.html>.
- Carlos Florensa, David Held, Markus Wulfmeier, Michael Zhang, and Pieter Abbeel. Reverse curriculum generation for reinforcement learning. In *Proceedings of the 1st Annual Conference on Robot Learning*, volume 78 of *Proceedings of Machine Learning Research*, pages 482–495, 2017. URL <https://proceedings.mlr.press/v78/florensa17a.html>.
- Alkis Gotovos, Nathalie Casati, Gregory Hitz, and Andreas Krause. Active learning for level set estimation. In *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence*, 2013. URL <https://people.csail.mit.edu/alkisg/files/gotovos13active.pdf>.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991. URL <https://arxiv.org/abs/1810.08240>.
- Minqi Jiang, Edward Grefenstette, and Tim Rocktäschel. Prioritized level replay. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4940–4950, 2021. URL <https://proceedings.mlr.press/v139/jiang21b.html>.
- Benjamin Letham, Phillip Guan, Chase Tymms, Eytan Bakshy, and Michael Shvartsman. Look-ahead acquisition functions for Bernoulli level set estimation. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 8493–8513, 2022. URL <https://proceedings.mlr.press/v151/letham22a.html>.
- Alex Rutherford, Michael Beukman, Timon Willi, Bruno Lacerda, Nick Hawes, and Jakob Foerster. No regrets: Investigating and improving regret approximations for curriculum discovery. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0512. URL https://papers.nips.cc/paper_files/paper/2024/hash/1d0ed12c3fda52f2c241a0cebcf739a6-Abstract-Conference.html.